

Daily Task 26/08/26

1. Create VPC with 2 private and 2 public subnets.

Objective: To create an AWS VPC with 2 public subnets and 2 private subnets.

VPC Configuration

VPC Name: DevOps-VPC

IPv4 CIDR Block: 172.168.0.0/24

AWS Region: US East (N. Virginia) – us-east-1

Tenancy: Default

IPv6 CIDR Block: Not Enabled

VPC Encryption Control: None

Subnet Configuration

Public Subnet 1

Subnet Name: public-subnet-1

Subnet Type: Public

IPv4 CIDR Block: 172.168.0.0/26

Availability Zone: us-east-1a

Public Subnet 2

Subnet Name: public-subnet-2

Subnet Type: Public

IPv4 CIDR Block: 172.168.0.64/26

Availability Zone: us-east-1b

Private Subnet 1

Subnet Name: private-subnet-1

Subnet Type: Private

IPv4 CIDR Block: 172.168.0.128/26

Availability Zone: us-east-1a

Private Subnet 2

Subnet Name: private-subnet-2

Subnet Type: Private

IPv4 CIDR Block: 172.168.0.192/26

Availability Zone: us-east-1b

Steps Performed

1. Opened the AWS Management Console and navigated to the VPC service.
2. Created a new VPC named DevOps-VPC.
3. Configured the IPv4 CIDR block as 172.168.0.0/24.
4. Created public-subnet-1 with CIDR 172.168.0.0/26 in us-east-1a.
5. Created public-subnet-2 with CIDR 172.168.0.64/26 in us-east-1b.

- Created private-subnet-1 with CIDR 172.168.0.128/26 in us-east-1a.
- Created private-subnet-2 with CIDR 172.168.0.192/26 in us-east-1b.
- Verified that all four subnets were successfully created inside DevOps-VPC.

Result

Successfully created an AWS VPC named DevOps-VPC with CIDR block 172.168.0.0/24 and configured 2 public subnets and 2 private subnets across two Availability Zones.

The first screenshot shows the AWS Management Console for the VPC 'vpc-05726b0885cfa4ae4 / DevOps-VPC'. The console displays a green success message: 'You successfully created vpc-05726b0885cfa4ae4 / DevOps-VPC'. The VPC details are as follows:

Details	Info
VPC ID	vpc-05726b0885cfa4ae4
DNS resolution	Enabled
Main network ACL	acl-0775a6df4805f15ef
IPv6 CIDR (Network border group)	-
Encryption control ID	-
State	Available
Tenancy	default
Default VPC	No
Network Address Usage metrics	Disabled
Encryption control mode	-
Block Public Access	Off
DHCP option set	dopt-0647ce68f2d9aa4e0
IPv4 CIDR	172.168.0.0/24
Route 53 Resolver DNS Firewall rule groups	-
DNS hostnames	Disabled
Main route table	rtb-02402781c765dc5be
IPv6 pool	-
Owner ID	634176595009

The second screenshot shows the AWS Management Console for the Subnets page. A green success message states: 'You have successfully created 1 subnet: subnet-080611d91b8f72fd2'. The Subnets table shows one subnet:

Name	Subnet ID	State	VPC
public-subnet-1	subnet-080611d91b8f72fd2	Available	vpc-05726b0885cfa4ae4 Dev...

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#subnets:subnetId=subnet-0848e97dbca782dd1

aws Search [Alt+S] United States (N. Virginia) CHVK (634176595009) CHVK

VPC > Subnets

VPC dashboard < Filter by VPC AWS Global View

▼ **Virtual private cloud**

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

Subnets (1) Info Last updated less than a minute ago Actions Create subnet

Find subnets by attribute or tag

Subnet ID : subnet-0848e97dbca782dd1 Clear filters

☐	Name	Subnet ID	State	VPC
☐	public-subnet-2	subnet-0848e97dbca782dd1	Available	vpc-05726b0885cfa4ae4 Dev...

Select a subnet

https://us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1 Mobile App © 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

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VPC > Subnets

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Your VPCs

Subnets

Route tables

Internet gateways

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DHCP option sets

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Managed prefix lists

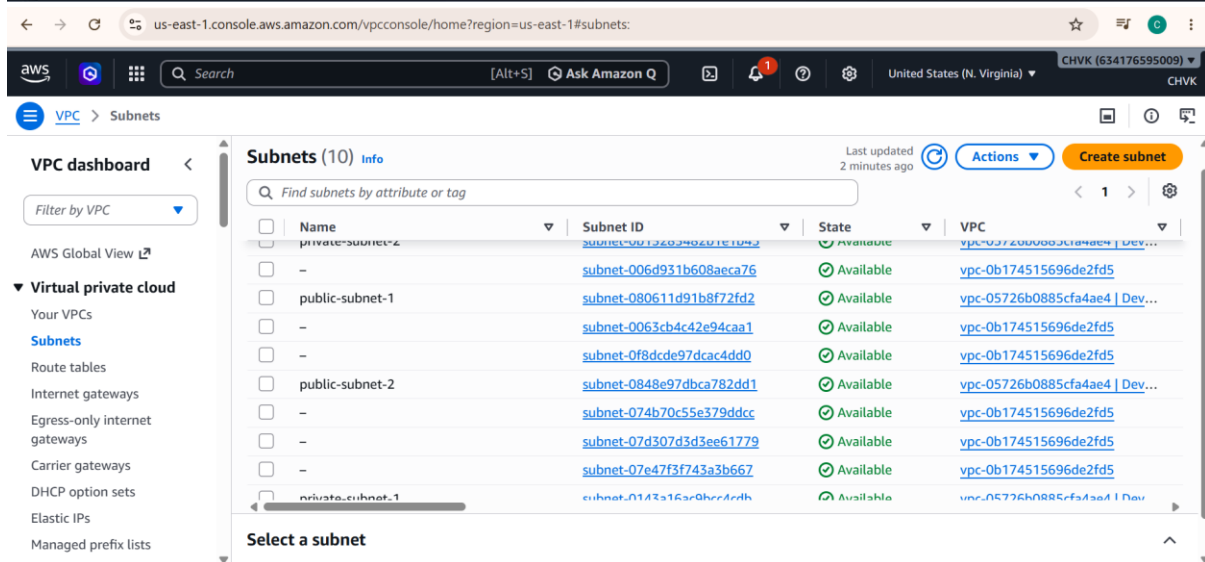
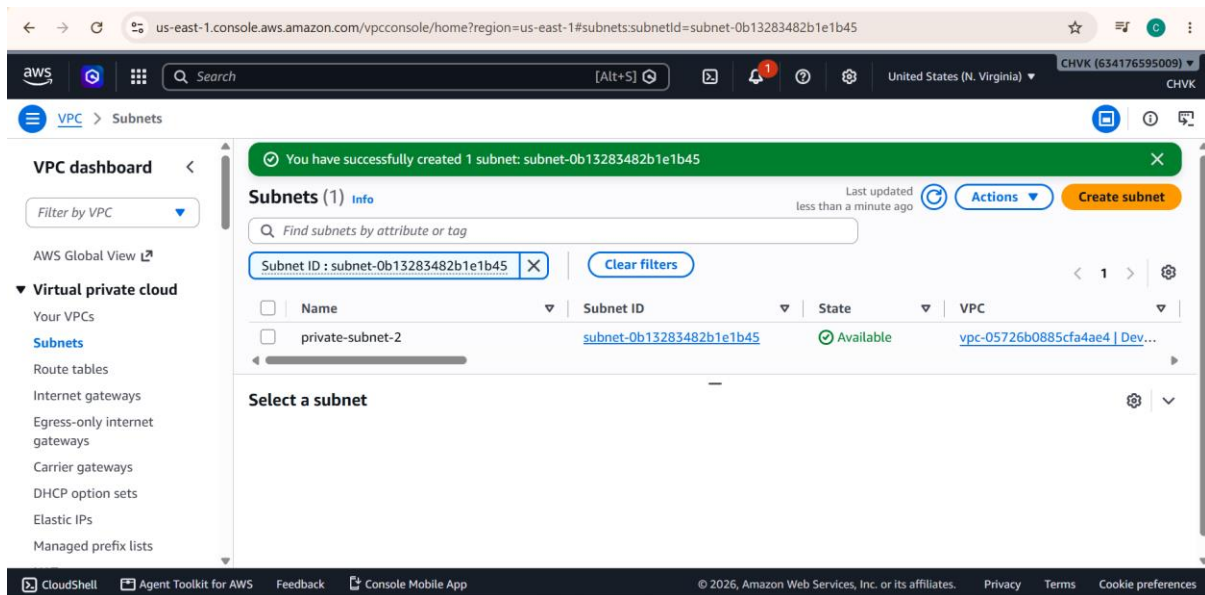
Subnets (1) Info Last updated less than a minute ago Actions Create subnet

Find subnets by attribute or tag

Subnet ID : subnet-0143a16ac9bcc4cdb Clear filters

☐	Name	Subnet ID	State	VPC
☐	private-subnet-1	subnet-0143a16ac9bcc4cdb	Available	vpc-05726b0885cfa4ae4 Dev...

Select a subnet



2. Enable DNS Hostname in VPC.

Objective : To enable DNS Hostnames for the DevOps-VPC.

VPC Configuration

VPC Name: DevOps-VPC

VPC CIDR Block: 172.168.0.0/24

AWS Region: US East (N. Virginia) – us-east-1

Steps Performed

Opened the AWS Management Console and navigated to the VPC service.

Navigated to Your VPCs.

Selected the VPC named DevOps-VPC.

Clicked Actions and selected Edit VPC settings.

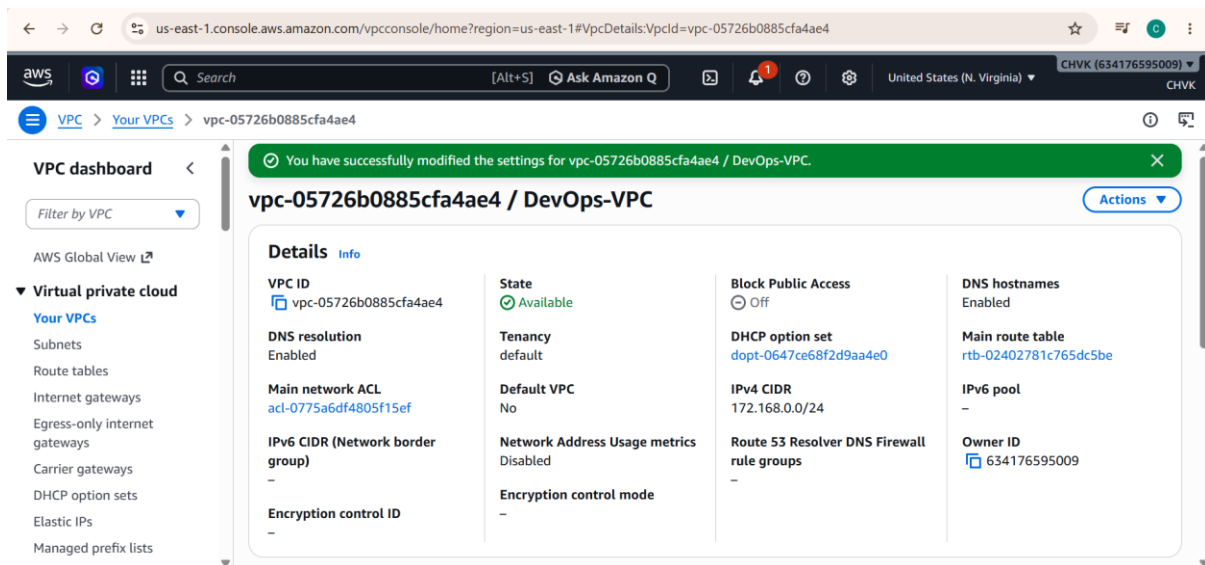
Under DNS settings, enabled Enable DNS hostnames.

Clicked Save changes.

Verified that DNS Hostnames was successfully enabled for the DevOps-VPC.

Result

Successfully enabled DNS Hostnames for the DevOps-VPC.



3. Enable Auto Assign Public IP in 2 public subnets.

Objective : To enable Auto-Assign Public IPv4 Address for both public subnets.

Subnet Configuration

Public Subnet 1

Subnet Name: public-subnet-1

CIDR Block: 172.168.0.0/26

Auto-Assign Public IPv4 Address: Yes

Public Subnet 2

Subnet Name: public-subnet-2

CIDR Block: 172.168.0.64/26

Auto-Assign Public IPv4 Address: Yes

Steps Performed

Opened the AWS Management Console and navigated to the VPC service.

Navigated to the Subnets section.

Selected public-subnet-1 and verified that Auto-Assign Public IPv4 Address was set to Yes.

Selected public-subnet-2 and verified that Auto-Assign Public IPv4 Address was set to Yes.

Confirmed that both public subnets are configured to automatically assign public IPv4 addresses to resources launched in them.

Evidence

Screenshot 1: public-subnet-1 showing Auto-Assign Public IPv4 Address as Yes.

Screenshot 2: public-subnet-2 showing Auto-Assign Public IPv4 Address as Yes.

Result

Successfully enabled and verified Auto-Assign Public IPv4 Address for both public subnets.

The image displays two screenshots of the AWS Management Console, specifically the 'Subnet Details' page for two public subnets. Both screenshots show a green notification banner at the top stating 'You have successfully changed subnet settings: Enable auto-assign public IPv4 address'. The left sidebar shows the 'VPC dashboard' with a filter set to 'VPC'. The main content area displays details for 'subnet-080611d91b8f72fd2 / public-subnet-1' and 'subnet-0848e97dbca782dd1 / public-subnet-2'. The details include Subnet ID, Subnet ARN, IPv4 CIDR, Availability Zone, Network ACL, Auto-assign customer-owned IPv4 address, Subnet ARN, Available IPv4 addresses, Network border group, Default subnet, Customer-owned IPv4 pool, State (Available), IPv6 CIDR, VPC (vpc-05726b0885cfa4ae4 | DevOps-VPC), Auto-assign public IPv4 address (Yes), Outpost ID, Block Public Access (Off), IPv6 CIDR association ID, Route table (rtb-02402781c765dc5be), Auto-assign IPv6 address (No), IPv4 CIDR reservations, and Resource name DNS A record.

4. Add 2 private subnets in private route table.

Objective : To create a private route table and associate both private subnets with it.

Route Table Configuration

Route Table Name: Private-Route-Table

VPC Name: DevOps-VPC

VPC CIDR Block: 172.168.0.0/24

Subnet Associations

Private Subnet 1: private-subnet-1

CIDR Block: 172.168.0.128/26

Private Subnet 2: private-subnet-2

CIDR Block: 172.168.0.192/26

Steps Performed

Opened the AWS Management Console and navigated to the VPC service.

Navigated to Route Tables.

Created a new route table named Private-Route-Table.

Selected DevOps-VPC as the associated VPC.

Opened the Subnet Associations tab.

Associated private-subnet-1 with the Private-Route-Table.

Associated private-subnet-2 with the Private-Route-Table.

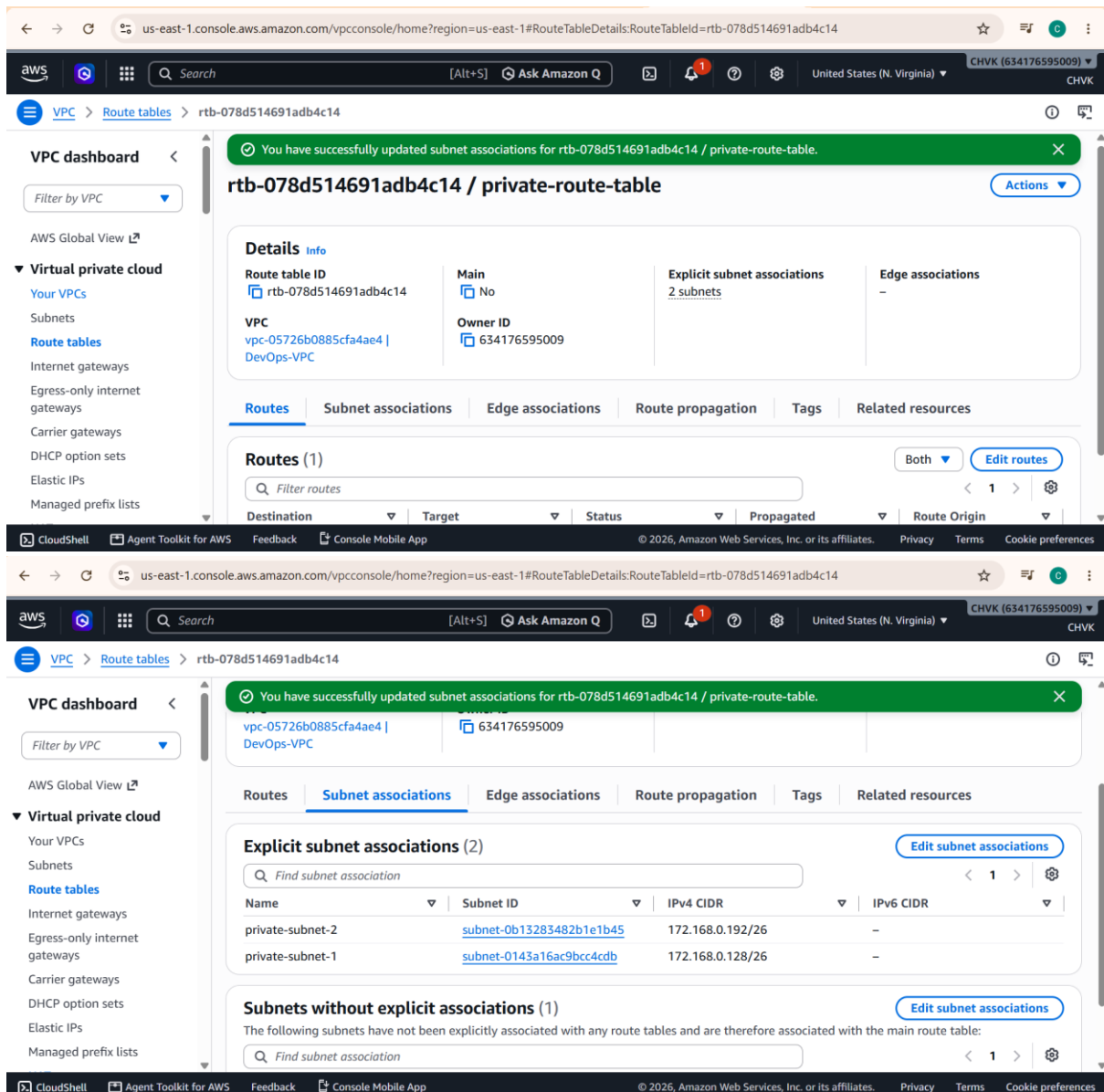
Verified that both private subnets were successfully associated with the private route table.

Result :Successfully created the Private-Route-Table and associated both private subnets with it.

The image displays two screenshots of the AWS Management Console, specifically the VPC dashboard, illustrating the steps to create and configure a private route table.

Top Screenshot: Shows the 'Route table rtb-078d514691adb4c14 / private-route-table' page. A green notification bar at the top states: 'Route table rtb-078d514691adb4c14 / private-route-table was created successfully.' The 'Details' section shows the route table ID, VPC (vpc-05726b0885cfa4ae4 | DevOps-VPC), and Owner ID. The 'Subnet associations' tab is selected, showing a list of routes (1) with columns for Destination, Target, Status, Propagated, and Route Origin. The 'Routes (1)' section shows a single route with Destination '0.0.0.0/0' and Target 'vpc-05726b0885cfa4ae4 | DevOps-VPC'.

Bottom Screenshot: Shows the same 'Route table rtb-078d514691adb4c14 / private-route-table' page. A green notification bar at the top states: 'You have successfully updated subnet associations for rtb-078d514691adb4c14 / private-route-table.' The 'Details' section shows the route table ID, VPC (vpc-05726b0885cfa4ae4 | DevOps-VPC), and Owner ID. The 'Subnet associations' tab is selected, showing a list of routes (1) with columns for Destination, Target, Status, Propagated, and Route Origin. The 'Routes (1)' section shows a single route with Destination '0.0.0.0/0' and Target 'vpc-05726b0885cfa4ae4 | DevOps-VPC'.



5. Add 2 public subnets in public route table.

Objective : To create a public route table and associate both public subnets with it.

Route Table Configuration

Route Table Name: Public-Route-Table

VPC Name: DevOps-VPC

VPC CIDR Block: 172.168.0.0/24

Subnet Associations

Public Subnet 1: public-subnet-1

CIDR Block: 172.168.0.0/26

Public Subnet 2: public-subnet-2

CIDR Block: 172.168.0.64/26

Steps Performed

Opened the AWS Management Console and navigated to the VPC service.

Navigated to Route Tables.

Created a new route table named Public-Route-Table.

Selected DevOps-VPC as the associated VPC.

Opened the Subnet Associations tab.

Associated public-subnet-1 with the Public-Route-Table.

Associated public-subnet-2 with the Public-Route-Table.

Verified that both public subnets were successfully associated with the Public-Route-Table.

Result

Successfully created the Public-Route-Table and associated both public subnets with it.

The first screenshot shows the AWS Management Console for the route table `rtb-00f3d9c6cef4ac955`. A green notification bar at the top states: "Route table rtb-00f3d9c6cef4ac955 | public-route-table was created successfully." The route table details show it is associated with VPC `vpc-05726b0885cfa4ae4` (DevOps-VPC) and has no explicit subnet associations or edge associations. The "Routes" tab shows one route.

The second screenshot shows the same route table after subnet associations have been added. A green notification bar states: "You have successfully updated subnet associations for rtb-00f3d9c6cef4ac955 / public-route-table." The "Subnet associations" tab is active, showing two explicit subnet associations:

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
public-subnet-1	subnet-080611d91b8f72fd2	172.168.0.0/26	-
public-subnet-2	subnet-0848e97dbca782dd1	172.168.0.64/26	-

Below this table, it shows "Subnets without explicit associations (0)".

6. Public route table will have the routes to internet and local.

Objective: To create an Internet Gateway, attach it to the DevOps-VPC, and configure an Internet route in the Public-Route-Table.

Internet Gateway Configuration

Internet Gateway Name: DevOps-IGW

VPC Name: DevOps-VPC

VPC CIDR Block: 172.168.0.0/24

Route Configuration

Route Table Name: Public-Route-Table

Destination: 0.0.0.0/0

Target: DevOps-IGW

Local Route: 172.168.0.0/24

Steps Performed

Opened the AWS Management Console and navigated to the VPC service.

Navigated to Internet Gateways.

Created an Internet Gateway named DevOps-IGW.

Attached the DevOps-IGW to the DevOps-VPC.

Navigated to Route Tables.

Selected the Public-Route-Table.

Opened the Routes tab and selected Edit routes.

Added a new route with destination 0.0.0.0/0.

Selected DevOps-IGW as the target for the Internet route.

Saved the route configuration.

Verified that the Public-Route-Table contains the local route and Internet Gateway route.

Result: Successfully configured the Internet Gateway and added an Internet route to the Public-Route-Table, providing Internet connectivity for the associated public subnets.

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#InternetGateway:internetGatewayId=igw-08fc71c73154e7d9e

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VPC > Internet gateways > igw-08fc71c73154e7d9e

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The following internet gateway was created: igw-08fc71c73154e7d9e - DevOps-IGW. You can now attach to a VPC to enable the VPC to communicate with the internet. **Attach to a VPC**

igw-08fc71c73154e7d9e / DevOps-IGW

Details Info

Internet gateway ID igw-08fc71c73154e7d9e	State Detached	VPC ID -	Owner 634176595009
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Tags (1) Manage tags

Key	Value
Name	DevOps-IGW

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us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#InternetGateway:internetGatewayId=igw-08fc71c73154e7d9e

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VPC > Internet gateways > igw-08fc71c73154e7d9e

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Internet gateway igw-08fc71c73154e7d9e successfully attached to vpc-05726b0885cfa4ae4

igw-08fc71c73154e7d9e / DevOps-IGW

Details Info

Internet gateway ID igw-08fc71c73154e7d9e	State Attached	VPC ID vpc-05726b0885cfa4ae4 DevOps-VPC	Owner 634176595009
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Tags (1) Manage tags

Key	Value
Name	DevOps-IGW

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us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#RouteTableDetails:RouteTableId=rtb-00f3d9c6cef4ac955

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VPC > Route tables > rtb-00f3d9c6cef4ac955

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Updated routes for rtb-00f3d9c6cef4ac955 / public-route-table successfully

Details

rtb-00f3d9c6cef4ac955	No	2 subnets	-
VPC vpc-05726b0885cfa4ae4 DevOps-VPC	Owner ID 634176595009		

Routes Subnet associations Edge associations Route propagation Tags Related resources

Routes (2) Edit routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-08fc71c73154e7d...	Active	No	Create Route
172.168.0.0/24	local	Active	No	Create Route Table

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7. Create EC2 in public subnet with t2.micro and install PHP.

Objective : To create an EC2 instance in the public subnet using the t2.micro instance type and install PHP.

EC2 Configuration

Instance Name: Public-PHP-Server

AMI: Ubuntu Server 24.04 LTS

Instance Type: t2.micro

VPC: DevOps-VPC

VPC CIDR Block: 172.168.0.0/24

Subnet: public-subnet-1

Availability Zone: us-east-1a

Auto-Assign Public IPv4: Enabled

Key Pair: ec2-task-key

Security Group: Public-PHP-SG

Security Group Configuration

SSH: TCP 22 – My IP

HTTP: TCP 80 – Anywhere IPv4 (0.0.0.0/0)

Steps Performed

Opened the AWS Management Console and navigated to the EC2 service.

Selected Launch Instance.

Selected Ubuntu Server 24.04 LTS as the AMI.

Selected t2.micro as the instance type.

Selected the existing ec2-task-key key pair.

Configured the network settings with DevOps-VPC and public-subnet-1.

Enabled Auto-Assign Public IPv4 Address.

Selected the existing Public-PHP-SG security group.

Launched the EC2 instance.

Verified that the EC2 instance was in the Running state.

Connected to the EC2 instance using SSH.

Updated the Ubuntu package repository using the following command:

```
sudo apt update
```

Installed PHP using the following command:

```
sudo apt install php -y
```

Verified the PHP installation using:

```
php -v
```

Result

Successfully created a public EC2 instance using t2.micro in public-subnet-1 and installed PHP successfully.

EC2 > Instances > i-081aefa2adc9be996

EC2

- Dashboard
- AWS Global View
- Events
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 - Instances
 - Instance Types
 - Launch Templates
 - Spot Requests
 - Savings Plans
 - Reserved Instances
 - Dedicated Hosts
 - Capacity Reservations
 - Capacity Manager
- ▼ Images

Instance summary for i-081aefa2adc9be996 (Public-PHP-Server) Info

Updated less than a minute ago

Instance ID
i-081aefa2adc9be996

IPv6 address
-

Hostname type
IP name: ip-172-168-0-45.ec2.internal

Answer private resource DNS name
-

Auto-assigned IP address
100.57.5.224 [Public IP]

Public IPv4 address
100.57.5.224 | [open address](#)

Instance state
Running

Private IP DNS name (IPv4 only)
ip-172-168-0-45.ec2.internal

Instance type
t2.micro

VPC ID
vpc-05726b0885cfa4ae4 (DevOps-)

Private IPv4 addresses
172.168.0.45

Public DNS
ec2-100-57-5-224.compute-1.amazonaws.com | [open address](#)

Elastic IP addresses
-

AWS Compute Optimizer finding
[Click to view details](#)

```
ubuntu@ip-172-168-0-45:~$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:5 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [971 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1229 kB]
Get:15 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [286 kB]
Get:16 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [181 kB]
Get:17 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [17.7 kB]
Get:18 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1688 kB]
Get:19 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [337 kB]
Get:20 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [388 kB]
Get:21 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [34.9 kB]
Get:22 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1486 kB]
Get:23 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [339 kB]
Get:24 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [45.4 kB]
Get:25 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [12.8 kB]
Get:26 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:27 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [656 B]
Get:28 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [40.6 kB]
Get:29 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main Translation-en [9172 B]
Get:30 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [5748 B]
Get:31 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [368 B]
Get:32 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [31.0 kB]
Get:33 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [18.6 kB]
Get:34 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [12.6 kB]
Get:35 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1588 B]
Get:36 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]
Get:37 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 c-n-f Metadata [116 B]
Get:38 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Packages [748 B]
Get:39 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse Translation-en [340 B]
Get:40 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Get:41 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 c-n-f Metadata [116 B]
Get:42 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [206 kB]
Get:43 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [46.4 kB]
Get:44 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1203 kB]
Get:45 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [240 kB]
```



```
ubuntu@ip-172-168-0-45:~$ sudo apt install php -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapache2-mod-php8.3 libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64 liblua5.4-0 php-common php8.3
  php8.3-cgi php8.3-common php8.3-opcache php8.3-readline ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser php-pear
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils libapache2-mod-php8.3 libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64 liblua5.4-0 php php-common
  php8.3 php8.3-cgi php8.3-common php8.3-opcache php8.3-readline ssl-cert
0 upgraded, 18 newly installed, 0 to remove and 101 not upgraded.
Need to get 7012 kB of archives.
After this operation, 30.5 MB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libapr1t64 amd64 1.7.2-3.1ubuntu0.1 [108 kB]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1t64 amd64 1.6.3-1.1ubuntu7 [91.9 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-1.1ubuntu7 [11.2 kB]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-ldap amd64 1.6.3-1.1ubuntu7 [9116 B]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 liblua5.4-0 amd64 5.4.6-3build2 [166 kB]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-bin amd64 2.4.58-1ubuntu8.15 [1338 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-data all 2.4.58-1ubuntu8.15 [163 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-utils amd64 2.4.58-1ubuntu8.15 [99.7 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2 amd64 2.4.58-1ubuntu8.15 [90.2 kB]
Get:10 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 php-common all 2:93ubuntu2 [13.9 kB]
Get:11 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-common amd64 8.3.6-0ubuntu0.24.04.10 [742 kB]
Get:12 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-opcache amd64 8.3.6-0ubuntu0.24.04.10 [372 kB]
Get:13 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-readline amd64 8.3.6-0ubuntu0.24.04.10 [13.5 kB]
Get:14 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3-cgi amd64 8.3.6-0ubuntu0.24.04.10 [1916 kB]
Get:15 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libapache2-mod-php8.3 amd64 8.3.6-0ubuntu0.24.04.10 [1848 kB]
Get:16 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 php8.3 all 8.3.6-0ubuntu0.24.04.10 [9172 B]
Get:17 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 php all 2:8.3+93ubuntu2 [4076 B]
Get:18 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 ssl-cert all 1.1.2ubuntu1 [17.8 kB]
Fetched 7012 kB in 0s (62.7 MB/s)
Preconfiguring packages ...
Selecting previously unselected package libapr1t64:amd64.
(Reading database ... 72608 files and directories currently installed.)
Preparing to unpack .../00-libapr1t64_1.7.2-3.1ubuntu0.1_amd64.deb ...
Unpacking libapr1t64:amd64 (1.7.2-3.1ubuntu0.1) ...
Selecting previously unselected package libaprutil1t64:amd64.
Preparing to unpack .../01-libaprutil1t64_1.6.3-1.1ubuntu7_amd64.deb ...
Unpacking libaprutil1t64:amd64 (1.6.3-1.1ubuntu7) ...
Selecting previously unselected package libaprutil1-dbd-sqlite3:amd64.
Preparing to unpack .../02-libaprutil1-dbd-sqlite3_1.6.3-1.1ubuntu7_amd64.deb ...
Unpacking libaprutil1-dbd-sqlite3:amd64 (1.6.3-1.1ubuntu7) ...
Selecting previously unselected package libaprutil1-ldap:amd64.
Preparing to unpack .../03-libaprutil1-ldap_1.6.3-1.1ubuntu7_amd64.deb ...
Unpacking libaprutil1-ldap:amd64 (1.6.3-1.1ubuntu7) ...
```

← → ↻ ⚠ Not secure 100.57.5.224/index.php ☆ ⚙ ⌵

PHP Version 8.3.6



System	Linux ip-172-168-0-45 6.17.0-1017-aws #17~24.04.1-Ubuntu SMP Tue May 26 21:30:32 UTC 2026 x86_64
Build Date	Jul 16 2026 18:30:41
Build System	Linux
Server API	Apache 2.0 Handler
Virtual Directory Support	disabled
Configuration File (php.ini) Path	/etc/php/8.3/apache2
Loaded Configuration File	/etc/php/8.3/apache2/php.ini
Scan this dir for additional .ini files	/etc/php/8.3/apache2/conf.d
Additional .ini files parsed	/etc/php/8.3/apache2/conf.d/10-opcache.ini, /etc/php/8.3/apache2/conf.d/10-pdo.ini, /etc/php/8.3/apache2/conf.d/20-calendar.ini, /etc/php/8.3/apache2/conf.d/20-ctype.ini, /etc/php/8.3/apache2/conf.d/20-exif.ini, /etc/php/8.3/apache2/conf.d/20-ffi.ini, /etc/php/8.3/apache2/conf.d/20-fileinfo.ini, /etc/php/8.3/apache2/conf.d/20-ftp.ini, /etc/php/8.3/apache2/conf.d/20-gettext.ini, /etc/php/8.3/apache2/conf.d/20-iconv.ini, /etc/php/8.3/apache2/conf.d/20-phar.ini, /etc/php/8.3/apache2/conf.d/20-posix.ini, /etc/php/8.3/apache2/conf.d/20-readline.ini, /etc/php/8.3/apache2/conf.d/20-shmop.ini, /etc/php/8.3/apache2/conf.d/20-sockets.ini, /etc/php/8.3/apache2/conf.d/20-sysmsg.ini, /etc/php/8.3/apache2/conf.d/20-sysvsem.ini, /etc/php/8.3/apache2/conf.d/20-sysvshm.ini, /etc/php/8.3/apache2/conf.d/20-tokenizer.ini
PHP API	20230831
PHP Extension	20230831
Zend Extension	420230831
Zend Extension Build	API420230831.NTS
PHP Extension Build	API20230831.NTS
Debug Build	no
Thread Safety	disabled

8. Configure NAT gateway in public subnet and connect to private instance.

Objective :To configure a NAT Gateway in the public subnet and provide internet access to the private subnets through the Private Route Table.

NAT Gateway Configuration

NAT Gateway Name: DevOps-NAT-Gateway

VPC: DevOps-VPC

Subnet: public-subnet-1

Connectivity Type: Public

Elastic IP Address: 18.204.146.116

NAT Gateway State: Available

Private Route Table Configuration

Route Table Name: private-route-table

VPC: DevOps-VPC

VPC CIDR Block: 172.168.0.0/24

Routes

Destination: 172.168.0.0/24

Target: local

Destination: 0.0.0.0/0

Target: NAT Gateway (DevOps-NAT-Gateway)

Steps Performed

Allocated an Elastic IP address from the Amazon IPv4 address pool.

Created a NAT Gateway named DevOps-NAT-Gateway.

Selected public-subnet-1 for the NAT Gateway.

Configured the NAT Gateway with Public connectivity.

Associated the allocated Elastic IP address with the NAT Gateway.

Waited until the NAT Gateway status changed to Available.

Navigated to VPC → Route Tables.

Selected the private-route-table.

Edited the existing 0.0.0.0/0 route.

Configured the target of the 0.0.0.0/0 route as the NAT Gateway.

Saved the route configuration successfully.

Verified that the private route table has two routes and two subnet associations.

Result

Successfully configured the NAT Gateway in the public subnet and configured the private route table to route internet-bound traffic through the NAT Gateway.

us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#Addresses:PublicIp=18.204.146.116

Search [Alt+S]

United States (N. Virginia) CHVK (634176595009) CHVK

VPC > Elastic IP addresses

Filter by VPC

AWS Global View

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections

Elastic IP address allocated successfully.
Elastic IP address 18.204.146.116

Associate this Elastic IP address

Elastic IP addresses (1) Info

Find elastic IP addresses by attribute or tag

Public IPv4 address : 18.204.146.116 Clear filters

	Name	Allocated IPv4 address	Type	Allocation ID
☐	-	18.204.146.116	Public IP	eipalloc-0268e38aaf9fa43d6

Select an elastic IP address

View IP address usage and recommendations to release unused IPs with [Public IP insights](#)

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us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#NatGatewayDetails:natGatewayId=nat-1b567ba69a66420be

Search [Alt+S]

United States (N. Virginia) CHVK (634176595009) CHVK

VPC > NAT gateways > nat-1b567ba69a66420be

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers

NAT gateway nat-1b567ba69a66420be | DevOps-NAT-Gateway was created successfully.

nat-1b567ba69a66420be / DevOps-NAT-Gateway

Details

NAT gateway ID nat-1b567ba69a66420be	Availability mode Regional	State Pending	State message Info -
NAT gateway ARN arn:aws:ec2:us-east-1:634176595009:natgateway/nat-1b567ba69a66420be	Connectivity type Public	Created Wednesday, August 26, 2026 at 17:27:47 GMT+5:30	Deleted -
VPC vpc-05726b0885cfa4ae4 / DevOps-VPC	Method of EIP allocation Automatic		

IP addresses Monitoring Flow logs Tags

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us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#RouteTableDetails:RouteTableId=rtb-078d514691adb4c14

Search [Alt+S]

United States (N. Virginia) CHVK (634176595009) CHVK

VPC > Route tables > rtb-078d514691adb4c14

Filter by VPC

AWS Global View

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways
- Peering connections
- Route servers

Updated routes for rtb-078d514691adb4c14 / private-route-table successfully

Details

rtb-078d514691adb4c14	No	2 subnets	-
VPC vpc-05726b0885cfa4ae4 DevOps-VPC	Owner ID 634176595009		

Routes Subnet associations Edge associations Route propagation Tags Related resources

Routes (2)

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	nat-1b567ba69a6642...	Active	No	Create Route
172.168.0.0/24	local	Active	No	Create Route Table

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9. Install Apache Tomcat in private EC2 and deploy a sample app.

Objective: To create an EC2 instance in a private subnet, install Apache Tomcat, and deploy a sample web application.

EC2 Configuration:

Instance Name: Private-Tomcat-Server

Instance ID: i-00b0acbd2616c3812

AMI: Ubuntu Server 24.04 LTS

Instance Type: t2.micro

VPC: DevOps-VPC

Subnet: private-subnet-1

Private IPv4 Address: 172.168.0.155

Public IPv4 Address: None

Key Pair: ec2-task-key

Steps Performed:

1. Created a private EC2 instance using Ubuntu Server 24.04 LTS.
2. Selected t2.micro as the instance type.
3. Placed the EC2 instance in private-subnet-1.
4. Disabled Auto-Assign Public IP.
5. Configured the security group to allow SSH (22) and Tomcat (8080) traffic from the Public-PHP-SG security group.
6. Connected to the private EC2 through the Public-PHP-Server.
7. Verified Java 17 installation using:
`java -version`
8. Downloaded Apache Tomcat 9.0.109 from the Apache archive.
9. Extracted Apache Tomcat and configured it under:
`/opt/tomcat`
10. Provided execute permissions to the Tomcat scripts using:
`sudo chmod +x /opt/tomcat/bin/*.sh`
11. Started Tomcat using:
`/opt/tomcat/bin/startup.sh`
12. Verified that Tomcat was running on port 8080.
13. Created a sample application directory:
`/opt/tomcat/webapps/sample`
14. Created an index.html file inside the sample application directory.
15. Tested the deployed application using:
`curl http://localhost:8080/sample/`

Result: Successfully created a private EC2 instance, installed Apache Tomcat, and deployed a sample application successfully.

EC2 > Instances

Instances (2) Info Last updated less than a minute ago Connect Instance state Actions Launch instances

Saved filter sets Running Find Instance by attribute or tag (case-sensitive)

Instance state = running Clear filters

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
☐	Public-PHP-Se...	i-081aefa2adc9be996	Running	t2.micro	2/2 checks pass	us-east-1a
☐	Private-Tomca...	i-00b0acbd2616c3812	Running	t2.micro	Initializing	us-east-1a

All instances

```
chvk@karthik:~/Downloads
$ ssh -i "ec2-task-key.pem" ubuntu@100.57.5.224
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1017-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Aug 26 12:24:20 UTC 2026

System load:  0.0      Processes:    120
Usage of /:   30.0% of 6.71GB      Users logged in: 1
Memory usage: 26%      IPv4 address for enx0: 172.168.0.45
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

103 updates can be applied immediately.
86 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

last login: Wed Aug 26 12:20:56 2026 from 103.160.27.199
ubuntu@ip-172-168-0-45:~$ ssh -i ec2-task-key.pem ubuntu@172.168.0.155
--r--r--r-- 1 ubuntu ubuntu 1678 Aug 26 12:24 ec2-task-key.pem
ubuntu@ip-172-168-0-45:~$ chmod 400 ec2-task-key.pem
ubuntu@ip-172-168-0-45:~$ ssh -i "ec2-task-key.pem" ubuntu@172.168.0.155
The authenticity of host '172.168.0.155 (172.168.0.155)' can't be established.
ED25519 key fingerprint is SHA256:EMagyx0XjpxfnHVxGbQCP2kp0dKoaFiuZrvkBNcc.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.168.0.155' (ED25519) to the list of known hosts.
Welcome to Ubuntu 24.04.4 LTS (GNU/Linux 6.17.0-1017-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Wed Aug 26 12:26:40 UTC 2026

System load:  0.08      Processes:    108
Usage of /:   26.6% of 6.71GB      Users logged in: 0
Memory usage: 23%      IPv4 address for enx0: 172.168.0.155
Swap usage:   0%
```

```
ubuntu@ip-172-168-0-155:~$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates InRelease [126 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports InRelease [126 kB]
Get:4 http://security.ubuntu.com/ubuntu noble-security InRelease [126 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Packages [15.0 MB]
Get:6 http://security.ubuntu.com/ubuntu noble-security/main amd64 Packages [971 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe Translation-en [5982 kB]
Get:8 http://security.ubuntu.com/ubuntu noble-security/main Translation-en [206 kB]
Get:9 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [46.4 kB]
Get:10 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [11.9 kB]
Get:11 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [1203 kB]
Get:12 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [240 kB]
Get:13 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [76.3 kB]
Get:14 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [24.2 kB]
Get:15 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1392 kB]
Get:16 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [321 kB]
Get:17 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [40.3 kB]
Get:18 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [11.1 kB]
Get:19 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [208 B]
Get:20 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [468 B]
Get:21 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 Components [3871 kB]
Get:22 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/universe amd64 c-n-f Metadata [301 kB]
Get:23 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Packages [269 kB]
Get:24 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse Translation-en [118 kB]
Get:25 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 Components [35.0 kB]
Get:26 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/multiverse amd64 c-n-f Metadata [8328 B]
Get:27 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Packages [1229 kB]
Get:28 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main Translation-en [12.8 kB]
Get:29 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 Components [181 kB]
Get:30 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 c-n-f Metadata [17.7 kB]
Get:31 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Packages [1688 kB]
Get:32 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe Translation-en [337 kB]
Get:33 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 Components [388 kB]
Get:34 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/universe amd64 c-n-f Metadata [34.9 kB]
Get:35 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted amd64 Packages [1486 kB]
Get:36 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/restricted Translation-en [339 kB]
Get:37 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Packages [45.4 kB]
Get:38 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse Translation-en [12.8 kB]
Get:39 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 Components [940 B]
Get:40 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/multiverse amd64 c-n-f Metadata [656 B]
Get:41 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 Packages [40.6 kB]
Get:42 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main Translation-en [9172 B]
Get:43 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 Components [5748 B]
Get:44 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [368 B]
Get:45 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [31.0 kB]
Get:46 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [18.6 kB]
```

```
ubuntu@ip-172-168-0-155:~$ sudo apt install openjdk-17-jdk -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  adwaita-icon-theme alsa-topology-conf alsa-ucm-conf at-spi2-common at-spi2-core ca-certificates-java dconf-gsettings-backend dconf-service fontconfig fontconfig-config
  fonts-dejavu-core fonts-dejavu-extra fonts-dejavu-mono gsettings-desktop-schemas gtk-update-icon-cache hicolor-icon-theme humanity-icon-theme java-common
  libasound2-data libasound2t64 libatk-bridge2.0-0t64 libatk-wrapper-java libatk-wrapper-java-jni libatk1.0-0t64 libatspi2.0-0t64 libbavahi-client3 libbavahi-common-data
  libbavahi-common3 libcairo-gobject2 libcairo2 libcups2t64 libdatrie1 libdeflate0 libdrm-amdgpu1 libdrm-common libdrm-intel1 libdrm2 libfontconfig1
  libgail-common3 libgail18t64 libgbm1 libgdk-pixbuf2.0-0 libgdk-pixbuf2.0-bin libgdk-pixbuf2.0-common libgif7 libgl1 libgl1-mesa-dri libglvnd0 libglx-mesa0 libglx0
  libgraphite2-3 libgtk2.0-0t64 libgtk2.0-bin libgtk2.0-common libharfbuzz0b libice-dev libice6 libjbig0 libjpeg-turbo8 libjpeg8 liblcms2-2 liblerc4 libllv20
  libpango-1.0-0 libpangocairo-1.0-0 libpangoft2-1.0-0 libpciaccess0 libpcsclite1 libpixman-1-0 libpthread-stubs0-dev librsync2-2 librsync2-common libsharpyuv0 libsm-dev
  libsm6 libthai-data libthai0 libtiff6 libvulkan1 libwayland-client0 libwebp7 libx11-dev libx11-xcb1 libxau-dev libxaw7 libxcb-dri3-0 libxcb-glx0 libxcb-present0
  libxcb-randr0 libxcb-render0 libxcb-shape0 libxcb-shm0 libxcb-sync1 libxcb-xfixes0 libxcb1-dev libxcomposite1 libxcursor1 libxdamage1 libxdmcp-dev libxfixes3 libxft2
  libxi6 libxinerama1 libxkbfile1 libxmu6 libxpm4 libxrandr2 libxrender1 libxshmfence1 libxt-dev libxt6t64 libxtst6 libxv1 libxxf86dga1 libxxf86vm1 mesa-libgallium
  mesa-vulkan-drivers openjdk-17-jdk-headless openjdk-17-jre openjdk-17-jre-headless session-migration ubuntu-mono x11-common x11-utils x11proto-dev xorg-sgml-doctools
  xtrans-dev
Suggested packages:
  default-jre alsa-utils libasound2-plugins cups-common gvfs libice-doc liblcms2-utils pcsd librsync2-bin libsm-doc libx11-doc libxcb-doc libxt-doc openjdk-17-demo
  openjdk-17-source visualvm libnss-mdns fonts-ipafont-gothic fonts-ipafont-mincho fonts-way-microhei | fonts-way-zenhei fonts-indic mesa-utils
Recommended packages:
  luit
The following NEW packages will be installed:
  adwaita-icon-theme alsa-topology-conf alsa-ucm-conf at-spi2-common at-spi2-core ca-certificates-java dconf-gsettings-backend dconf-service fontconfig fontconfig-config
  fonts-dejavu-core fonts-dejavu-extra fonts-dejavu-mono gsettings-desktop-schemas gtk-update-icon-cache hicolor-icon-theme humanity-icon-theme java-common
  libasound2-data libasound2t64 libatk-bridge2.0-0t64 libatk-wrapper-java libatk-wrapper-java-jni libatk1.0-0t64 libatspi2.0-0t64 libbavahi-client3 libbavahi-common-data
  libbavahi-common3 libcairo-gobject2 libcairo2 libcups2t64 libdatrie1 libdeflate0 libdrm-amdgpu1 libdrm-common libdrm-intel1 libdrm2 libfontconfig1 libgail-common3
  libgail18t64 libgbm1 libgdk-pixbuf2.0-0 libgdk-pixbuf2.0-bin libgdk-pixbuf2.0-common libgif7 libgl1 libgl1-mesa-dri libglvnd0 libglx-mesa0 libglx0 libgraphite2-3
  libgtk2.0-0t64 libgtk2.0-bin libgtk2.0-common libharfbuzz0b libice-dev libice6 libjbig0 libjpeg-turbo8 libjpeg8 liblcms2-2 liblerc4 libllv20 libpango-1.0-0
  libpangocairo-1.0-0 libpangoft2-1.0-0 libpciaccess0 libpcsclite1 libpixman-1-0 libpthread-stubs0-dev librsync2-2 librsync2-common libsharpyuv0 libsm-dev libsm6
  libthai-data libthai0 libtiff6 libvulkan1 libwayland-client0 libwebp7 libx11-dev libx11-xcb1 libxau-dev libxaw7 libxcb-dri3-0 libxcb-glx0 libxcb-present0 libxcb-render0
  libxcb-shape0 libxcb-shm0 libxcb-sync1 libxcb-xfixes0 libxcb1-dev libxcomposite1 libxcursor1 libxdamage1 libxdmcp-dev libxfixes3 libxft2 libxi6 libxinerama1 libxkbfile1
  libxmu6 libxpm4 libxrandr2 libxrender1 libxshmfence1 libxt-dev libxt6t64 libxtst6 libxv1 libxxf86dga1 libxxf86vm1 mesa-libgallium mesa-vulkan-drivers openjdk-17-jdk
  openjdk-17-jdk-headless openjdk-17-jre openjdk-17-jre-headless session-migration ubuntu-mono x11-common x11-utils x11proto-dev xorg-sgml-doctools xtrans-dev
The following packages will be upgraded:
  libdrm-amdgpu1 libdrm-common libdrm2
3 upgraded, 126 newly installed, 0 to remove and 98 not upgraded.
Need to get 189 MB of archives.
After this operation, 655 MB of additional disk space will be used.
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libdrm-common all 2.4.125-1ubuntu0.1-24.04.2 [9250 B]
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libdrm2 amd64 2.4.125-1ubuntu0.1-24.04.2 [41.4 kB]
Get:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libgdk-pixbuf2.0-common all 2.42.10+dfsg-3ubuntu3.3 [8302 B]
Get:4 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libjpeg-turbo8 amd64 2.1.5-2ubuntu2 [150 kB]
Get:5 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libjpeg8 amd64 8c-2ubuntu1 [2148 B]
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libdeflate0 amd64 1.19-1build1 [43.9 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libjbig0 amd64 2.1-6.ubuntu2 [29.7 kB]
Get:8 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 liblerc4 amd64 4.0.0-4ubuntu2 [179 kB]
Get:9 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libsharpyuv0 amd64 1.3.2-0.4build3 [15.8 kB]

ubuntu@ip-172-168-0-155:~$ wget https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.109/bin/apache-tomcat-9.0.109.tar.gz
--2026-08-26 12:32:48-- https://archive.apache.org/dist/tomcat/tomcat-9/v9.0.109/bin/apache-tomcat-9.0.109.tar.gz
Resolving archive.apache.org (archive.apache.org)... 65.108.204.189, 2a01:4f9:1a:a084::2
Connecting to archive.apache.org (archive.apache.org)|65.108.204.189|:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 13035397 (12M) [application/x-gzip]
Saving to: 'apache-tomcat-9.0.109.tar.gz'

apache-tomcat-9.0.109.tar.gz 100%[=====] 12.43M 264KB/s in 62s

2026-08-26 12:33:51 (204 KB/s) - 'apache-tomcat-9.0.109.tar.gz' saved [13035397/13035397]

ubuntu@ip-172-168-0-155:~$ tmp$ ls -lh apache-tomcat-9.0.109.tar.gz
-rw-r--r-- 1 ubuntu ubuntu 13M Sep 1 2025 apache-tomcat-9.0.109.tar.gz
ubuntu@ip-172-168-0-155:~$ tmp$ tar -xzf apache-tomcat-9.0.109.tar.gz
ubuntu@ip-172-168-0-155:~$ tmp$ sudo mv apache-tomcat-9.0.109/* /opt/tomcat
ubuntu@ip-172-168-0-155:~$ tmp$ sudo chmod +x /opt/tomcat/bin/*.sh
ubuntu@ip-172-168-0-155:~$ tmp$ /opt/tomcat/bin/startup.sh
Using CATALINA_BASE: /opt/tomcat
Using CATALINA_HOME: /opt/tomcat
Using CATALINA_TMPDIR: /opt/tomcat/temp
Using JRE_HOME: /usr
Using CLASSPATH: /opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar
Tomcat started.
ubuntu@ip-172-168-0-155:~$ tmp$ ps -ef | grep tomcat
ubuntu 3489 1 6 12:35 pts/0 00:00:02 /usr/bin/java -Djava.util.logging.config.file=/opt/tomcat/conf/logging.properties -Djava.util.logging.manager=org.apache
catalina.classloader.DefaultManager -Djdk.tls.ephemeralDHKeySize=2048 -Djava.protocol.handler.pkgs=org.apache.catalina.webresources -Dsun.io.useCanonicalCaches=false -Dorg.apache.cata
lina.security.SecurityListener.UMASK=0027 -Dignore.endorsed.dirs=-classpath:/opt/tomcat/bin/bootstrap.jar:/opt/tomcat/bin/tomcat-juli.jar -Dcatalina.base=/opt/tomcat -Dcat
lina.home=/opt/tomcat -Djava.io.tmpdir=/opt/tomcat/temp org.apache.catalina.startup.Bootstrap start
ubuntu 3520 1129 0 12:36 pts/0 00:00:00 grep --color=auto tomcat
ubuntu@ip-172-168-0-155:~$ tmp$ ss -tulnp | grep 8080
tcp LISTEN 0 100 *:* 0.0.0.0:8080 *:* users:((("java",pid=3489,fd=44))
ubuntu@ip-172-168-0-155:~$ tmp$ curl http://localhost:8080

<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="UTF-8" />
<title>Apache Tomcat/9.0.109</title>
<link href="favicon.ico" rel="icon" type="image/x-icon" />
<link href="tomcat.css" rel="stylesheet" type="text/css" />
</head>
<body>
<div id="wrapper">
```



```

</head>
<body>
  <div id="wrapper">
    <div id="navigation" class="curved container">
      <span id="nav-home"><a href="https://tomcat.apache.org/">Home</a></span>
      <span id="nav-hosts"><a href="/docs/">Documentation</a></span>
      <span id="nav-config"><a href="/docs/config/">Configuration</a></span>
      <span id="nav-examples"><a href="/examples/">Examples</a></span>
      <span id="nav-wiki"><a href="https://cwiki.apache.org/confluence/display/TOMCAT/">Wiki</a></span>
      <span id="nav-lists"><a href="https://tomcat.apache.org/lists.html">Mailing Lists</a></span>
      <span id="nav-help"><a href="https://tomcat.apache.org/findhelp.html">Find Help</a></span>
      <br class="separator" />
    </div>
    <div id="asf-box">
      <h1>Apache Tomcat/9.0.109</h1>
    </div>
    <div id="upper" class="curved container">
      <div id="congrats" class="curved container">
        <h2>If you're seeing this, you've successfully installed Tomcat. Congratulations!</h2>
      </div>
      <div id="notice">
        
        <div id="tasks">
          <h3>Recommended Reading:</h3>
          <h4><a href="/docs/security-howto.html">Security Considerations How-To</a></h4>
          <h4><a href="/docs/manager-howto.html">Manager Application How-To</a></h4>
          <h4><a href="/docs/cluster-howto.html">Clustering/Session Replication How-To</a></h4>
        </div>
      </div>
      <div id="actions">
        <div class="button">
          <a class="container shadow" href="/manager/status"><span>Server Status</span></a>
        </div>
        <div class="button">
          <a class="container shadow" href="/manager/html"><span>Manager App</span></a>
        </div>
        <div class="button">
          <a class="container shadow" href="/host-manager/html"><span>Host Manager</span></a>
        </div>
      </div>
      <br class="separator" />
    </div>
    <div id="middle" class="curved container">
      <h3>Developer Quick Start</h3>
      <div class="col25">
        <div class="container">
          <p><a href="/docs/setup.html">Tomcat Setup</a></p>
          <ul>
            <li><a href="https://tomcat.apache.org/connectors-doc/">Tomcat Connectors</a></li>
            <li><a href="https://tomcat.apache.org/connectors-doc/">mod_jk Documentation</a></li>
            <li><a href="https://tomcat.apache.org/native-doc/">Tomcat Native</a></li>
            <li><a href="/docs/deployer-howto.html">Deployer</a></li>
          </ul>
        </div>
      </div>
      <div class="col20">
        <div class="container">
          <h4>Get Involved</h4>
          <ul>
            <li><a href="https://tomcat.apache.org/getinvolved.html">Overview</a></li>
            <li><a href="https://tomcat.apache.org/source.html">Source Repositories</a></li>
            <li><a href="https://tomcat.apache.org/lists.html">Mailing Lists</a></li>
            <li><a href="https://cwiki.apache.org/confluence/display/TOMCAT/">Wiki</a></li>
          </ul>
        </div>
      </div>
      <div class="col20">
        <div class="container">
          <h4>Miscellaneous</h4>
          <ul>
            <li><a href="https://tomcat.apache.org/contact.html">Contact</a></li>
            <li><a href="https://tomcat.apache.org/legal.html">Legal</a></li>
            <li><a href="https://www.apache.org/foundation/sponsorship.html">Sponsorship</a></li>
            <li><a href="https://www.apache.org/foundation/thanks.html">Thanks</a></li>
          </ul>
        </div>
      </div>
      <div class="col20">
        <div class="container">
          <h4>Apache Software Foundation</h4>
          <ul>
            <li><a href="https://tomcat.apache.org/whoweare.html">Who We Are</a></li>
            <li><a href="https://tomcat.apache.org/heritage.html">Heritage</a></li>
            <li><a href="https://www.apache.org">Apache Home</a></li>
            <li><a href="https://tomcat.apache.org/resources.html">Resources</a></li>
          </ul>
        </div>
      </div>
      <br class="separator" />
      <p class="copyright">Copyright &copy;1999-2026 Apache Software Foundation. All Rights Reserved</p>
    </div>
  </body>
</html>
ubuntu@ip-172-168-0-155: /tmp$

```

The screenshot shows the Apache Tomcat 9.0.109 web interface in a browser at localhost:8080. The page has a navigation bar with links: Home, Documentation, Configuration, Examples, Wiki, Mailing Lists, and a Find Help button. The main heading is "Apache Tomcat/9.0.109". Below it, a green banner says "If you're seeing this, you've successfully installed Tomcat. Congratulations!". To the left is the Tomcat logo (a cat). To the right, under "Recommended Reading:", are links for "Security Considerations How-To", "Manager Application How-To", and "Clustering/Session Replication How-To". Further right are buttons for "Server Status", "Manager App", and "Host Manager". Below this is a "Developer Quick Start" section with links for "Tomcat Setup", "First Web Application", "Realms & AAA", "JDBC DataSources", "Examples", "Servlet Specifications", and "Tomcat Versions". At the bottom, there are three yellow boxes: "Managing Tomcat" (with security warnings and a link to Read more), "Documentation" (with links to Tomcat 9.0 Documentation, Tomcat 9.0 Configuration, and Tomcat Wiki), and "Getting Help" (with links to FAQ and Mailing Lists, and a list of mailing lists: tomcat-announce, tomcat-users). Below the web interface is a terminal window showing the following commands and output:

```
ubuntu@ip-172-168-0-155:~$ cd /opt/tomcat/webapps
ubuntu@ip-172-168-0-155:/opt/tomcat/webapps$ ls
ROOT docs examples host-manager manager
ubuntu@ip-172-168-0-155:/opt/tomcat/webapps$ sudo mkdir sample
ubuntu@ip-172-168-0-155:/opt/tomcat/webapps$ sudo nano sample/index.html
ubuntu@ip-172-168-0-155:/opt/tomcat/webapps$ curl http://localhost:8080/sample/
<html>
<head>
<title>Sample Application</title>
</head>
<body>
<h1>Welcome to My Sample Application</h1>
<p>Deployed on Apache Tomcat Private EC2</p>
</body>
</html>
ubuntu@ip-172-168-0-155:/opt/tomcat/webapps$
```

10. Configure VPC flow logs and store the logs in S3 and CloudWatch.

Objective:

To enable VPC Flow Logs for the DevOps-VPC and configure the logs to be delivered to Amazon S3 and Amazon CloudWatch Logs.

VPC Configuration:

VPC Name: DevOps-VPC

VPC ID: vpc-05726b0885cfa4ae4

Region: US East (N. Virginia)

PART 1 – VPC FLOW LOGS TO AMAZON S3

Steps Performed:

1. Opened the AWS VPC Console.

2. Selected the DevOps-VPC.

3. Navigated to the Flow Logs section.

4. Created an S3 bucket for storing VPC Flow Logs.

S3 Bucket Name:

devops-vpc-flowlogs-chvk-2026

5. Created a VPC Flow Log with the following configuration:

Flow Log Name:

DevOps-VPC-S3-Flow-Logs

Traffic Type:

All

Maximum Aggregation Interval:

1 minute

Destination:

Amazon S3

S3 Bucket ARN:

arn:aws:s3:::devops-vpc-flowlogs-chvk-2026

Log Record Format:

AWS Default Format

Log File Format:

Text

6. Successfully created the S3 Flow Log and verified that its status was Active.

PART 2 – VPC FLOW LOGS TO CLOUDWATCH

Steps Performed:

1. Created a CloudWatch Log Group:

/aws/vpc/flowlogs

2. Created an IAM role for VPC Flow Logs to publish logs to CloudWatch Logs.

IAM Role Name:

VPCFlowLogsCloudWatchRole

3. Created another VPC Flow Log with the following configuration:

Flow Log Name:

DevOps-VPC-CloudWatch-Flow_Logs

Traffic Type:

All

Maximum Aggregation Interval:

1 minute

Destination:

CloudWatch Logs

Destination Log Group:

/aws/vpc/flowlogs

IAM Role:

VPCFlowLogsCloudWatchRole

Log Record Format:

AWS Default Format

4. Successfully created the CloudWatch Flow Log.

5. Verified that the Flow Log status is Active.

RESULT:

VPC Flow Logs were successfully configured for DevOps-VPC.

Logs are configured to be delivered to:

1. Amazon S3
2. Amazon CloudWatch Logs

Both Flow Log configurations were successfully created and verified.

devops-vpc-flowlogs-chvk-2026 [Info](#)

Objects Metadata **Properties** Permissions Metrics Management File systems - new Access Points

Bucket overview

AWS Region

US East (N. Virginia) us-east-1

✓ Bucket ARN copied

Amazon resource Name (ARN)

arn:aws:s3::devops-vpc-flowlogs-chvk-2026

Creation date

August 26, 2026, 18:57:24 (UTC+05:30)

Bucket Versioning [Edit](#)

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in your Amazon S3 bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

Disabled

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

✓ Successfully created flow log for the following resource:

vpc-05726b0885cfa4ae4

[Explore integration options](#)

fl-0097fc2aea989d473 / DevOps-VPC-Flow-Logs [Actions](#)

Details

Flow Log ID

fl-0097fc2aea989d473

Name

DevOps-VPC-Flow-Logs

State

Active

Creation Time

Wednesday, August 26, 2026 at 19:02:11 GMT+5:30

Destination Type

s3

Destination Name

[devops-vpc-flowlogs-chvk-2026](#)

IAM Role

-

Cross Account IAM Role

-

Traffic Type

All

Max Aggregation Interval

1 minute

Log Format

Default

Tag field specifications

-

File Format

Plain text

Hive Compatible Partitions

Not enabled

Partition Logs

Daily

VPC dashboard

Filter by VPC

AWS Global View

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

Carrier gateways

DHCP option sets

Elastic IPs

Managed prefix lists

✓ Successfully created flow log for the following resource:

vpc-05726b0885cfa4ae4

fl-06e6f5a379452197b / DevOps-VPC-CloudWatch-Flow_Logs [Actions](#)

Details

Flow Log ID

fl-06e6f5a379452197b

Name

DevOps-VPC-CloudWatch-Flow_Logs

State

Active

Creation Time

Wednesday, August 26, 2026 at 19:15:40 GMT+5:30

Destination Type

cloud-watch-logs

Destination Name

[aws/vpc/flowlogs](#)

IAM Role

[arn:aws:iam::634176595009:role/VPCFlowLogsCloudWatchRole](#)

Cross Account IAM Role

-

Traffic Type

All

Max Aggregation Interval

1 minute

Log Format

Default

Tag field specifications

-

File Format

-

Hive Compatible Partitions

-

Partition Logs

-